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Embedded System Design

CASE STUDY REPORT

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1. Executive Summary

The Smart Hostel Electricity Saver Auto Cut-Off System is an Embedded Systems Design (ESD) project aimed at reducing unnecessary electricity consumption in hostel rooms. Electrical appliances such as lights, fans, and chargers are often left switched on even when rooms are unoccupied, leading to energy wastage. The core problem is the absence of an embedded control system that can automatically manage electrical loads based on real-time occupancy. Manual control is inefficient and unreliable. The proposed solution is a sensor-based embedded system that uses motion/presence sensors, a microcontroller, and relay modules. The system continuously monitors room occupancy and automatically disconnects power when no presence is detected for a specified duration. The system is expected to improve energy efficiency, reduce electricity wastage, and demonstrate the practical application of embedded systems in real-world scenarios.

2. Introduction

Background of the Project

Electricity wastage in hostels is a common issue due to manual control of appliances. Embedded systems provide a reliable way to automate such processes by integrating sensing, processing, and control mechanisms. This project uses an embedded system to automatically control electricity based on room occupancy, eliminating dependency on manual switching.

Technical Context

Embedded systems are widely used in automation, control systems, and smart infrastructure. They combine hardware (microcontrollers, sensors, actuators) and software (firmware) to perform dedicated real-time tasks efficiently.

This project applies embedded system concepts such as:

- Real-time sensing
- Control logic execution
- Hardware–software interfacing
- Low-power system design

Purpose of the Case Study

The purpose of this project is to design and analyze a smart automatic electricity cut off system that can help reduce power wastage in hostel rooms. The system uses sensors to detect the presence of occupants and automatically controls electrical supply based on room occupancy. This case study examines how such a system can contribute to energy efficiency, cost reduction, and sustainable energy usage in hostel.

Scope and Limitations

Scope

- Design of a microcontroller-based automation system
- Integration of PIR sensor and relay module
- Real-time monitoring and control.

Limitations

- Sensor may fail to detect stationary occupants
 - Performance depends on sensor placement
 - Limited to prototype-scale implementation
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3. Problem Statement

Clear Definition of the Core Issue

In many hostels, electrical appliances such as lights, fans, and chargers are often left switched on even when rooms are unoccupied. This happens because students frequently forget to turn off these devices when leaving their rooms. As a result, electricity continues to be consumed unnecessarily for long periods. The core issue is the lack of an automated system that can detect room occupancy and automatically control the electrical supply to prevent energy wastage.

Why It Is Important

Energy conservation is an important concern due to increasing electricity demand and rising energy costs. Hostels typically have a large number of rooms, and even small amounts of wasted electricity in each room can accumulate into significant energy loss. Implementing a smart automation system that automatically cuts off power when the room is empty can help

promote efficient energy usage, reduce electricity consumption, and encourage sustainable practices.

Impact on the Organization

The absence of an automatic electricity control system can lead to several negative impacts on hostel management, including:

- Higher electricity bills due to continuous power consumption.
- Increased energy wastage, which affects sustainability efforts.
- Higher operational costs for maintaining hostel facilities.
- Difficulty in monitoring electricity usage across multiple rooms.

By introducing a Smart Hostel Electricity Saver Auto Cut-Off System, hostels can significantly reduce unnecessary power consumption, lower operational costs, and improve overall energy management.

4. Situational Analysis

A. Internal Analysis

Internal analysis examines the internal strengths, weaknesses, resources, and capabilities of the Smart Hostel Electricity Saver Auto Cut-Off System project.

Strengths

- **Energy Efficiency:** The system helps reduce unnecessary electricity consumption in hostel rooms.
- **Automation:** Automatically switches off electrical appliances when no person is detected.
- **Low-Cost Implementation:** Uses affordable components such as sensors, relays, and microcontrollers.
- **Easy Installation:** The system can be integrated into existing electrical setups in hostels.
- **Environmentally Friendly:** Supports energy conservation and sustainability.

Weaknesses

- **Sensor Dependency:** The system relies heavily on the accuracy of motion or presence sensors.
- **Limited Detection:** If a person remains completely still, the system may incorrectly assume the room is empty.
- **Initial Setup Cost:** Although low, some initial investment is required for installation.
- **Maintenance Requirements:** Sensors and electronic components may require periodic maintenance.

Resources and Capabilities (SWOT / VRIO Perspective)

Resources

- Microcontroller platforms (such as Arduino)
- Motion sensors (PIR sensors)
- Relay modules for controlling electrical supply
- Basic programming and embedded system knowledge

Capabilities

- Ability to automate power management in rooms
- Real-time detection of room occupancy
- Integration of hardware and software for smart control

From a VRIO perspective, the project's value lies in its cost-effective automation and energy saving capability, which can provide practical benefits for hostels and similar residential environments.

B. External Analysis

External analysis examines external factors that can influence the success or adoption of the system.

Opportunities

- Increasing demand for smart energy management systems.
- Rising electricity costs encouraging institutions to reduce energy consumption.
- Growth of smart building and IoT technologies.

- Government initiatives promoting energy conservation and sustainability.

Threats

- Availability of advanced commercial smart home automation systems.
- Competition from established automation and energy management solutions.
- Technical issues such as sensor malfunction or power fluctuations.
- Resistance to adopting new technologies due to installation costs.

Industry Forces (PESTLE and Porter's Five Forces)

PESTLE Analysis

- Political: Government policies promoting energy efficiency and sustainable energy usage.
- Economic: Rising electricity costs increase demand for energy-saving solutions.
- Social: Growing awareness about environmental sustainability and responsible energy usage.
- Technological: Rapid development of IoT, sensors, and smart automation technologies.
- Legal: Regulations related to energy consumption and building safety standards.
- Environmental: Need to reduce carbon footprint and energy wastage.

Porter's Five Forces

- Competitive Rivalry: Moderate to high due to existing smart automation solutions.
- Threat of New Entrants: Moderate because technology components are widely available.
- Bargaining Power of Buyers: High since customers can choose from multiple energy management solutions.
- Bargaining Power of Suppliers: Low because electronic components are easily available.
- Threat of Substitutes: Moderate due to alternative smart home automation systems.

This situational analysis shows that the Smart Hostel Electricity Saver Auto Cut-Off System has strong potential due to growing demand for energy-efficient technologies, although it must compete with other automation solutions in the market.

5. Alternatives / Options

Option 1: Manual Electricity Control System

This approach relies on students manually switching off lights and appliances before leaving the room.

Pros

- No installation cost
- Simple and easy to implement
- No technical maintenance required

Cons

- Highly dependent on human behavior
- High chances of electricity wastage
- No monitoring or automation

Risk Assessment

- High risk of continued energy wastage due to forgetfulness or negligence.
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Option 2: Timer-Based Automatic System

This system automatically switches off electrical appliances after a preset time interval.

Pros

- Reduces electricity wastage to some extent
- Easy to implement using timer circuits
- Low-cost solution

Cons

- Does not detect real-time occupancy
- Appliances may turn off while someone is still in the room
- Limited flexibility

Risk Assessment

- Medium risk because fixed timers may not match real usage patterns.

Option 3: Smart Sensor-Based Auto Cut-Off System (Proposed System)

This system uses motion or presence sensors and a microcontroller to detect whether someone is in the room and automatically control electricity.

Pros

- Real-time detection of room occupancy
- Reduces electricity wastage effectively
- Fully automated system
- Energy efficient and scalable

Cons

- Slightly higher installation cost
- Requires sensors and microcontroller programming
- Possible sensor inaccuracies in rare cases

Risk Assessment

- Low risk compared to other options, as the system provides automated and intelligent energy control.

6. Recommendation

Selected Solution

The Smart Sensor-Based Auto Cut-Off System is recommended as the best solution.

Justification

- Provides automatic electricity control based on room occupancy.
- Reduces electricity wastage significantly compared to manual or timer-based systems.
- Uses affordable components such as sensors, relays, and microcontrollers, making it cost-effective.
- Improves overall energy efficiency and sustainability.

Alignment with Organizational Goals

The system aligns with hostel management goals by:

- Reducing electricity bills
 - Promoting energy conservation
 - Improving operational efficiency
 - Supporting environmentally sustainable practices
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7. Implementation Plan

Action Steps

1. Identify rooms where the system will be installed.
2. Procure required hardware components (sensors, microcontrollers, relays).
3. Design and assemble the circuit system.
4. Program the microcontroller for sensor detection and power control.
5. Install the system in selected hostel rooms.
6. Test the system and monitor its performance.
7. Expand implementation to additional rooms if successful.

Timeline

Phase	Duration
Planning & Design	1 week
Component Procurement	1 week
System Development	2 weeks
Testing & Implementation	1 week
Total estimated time: 5 weeks	

Required Resources

- Microcontroller (Arduino or similar)
- PIR motion sensors
- Relay modules
- Power supply unit
- Electrical wiring components
- Programming software

Budget Considerations

Component	Estimated Cost
Microcontroller	₹500 – ₹800
PIR Sensor	₹150 – ₹300
Relay Module	₹200 – ₹400
Wiring & Miscellaneous	₹200 – ₹300
Estimated cost per room: ₹1000 – ₹1800	

Key Stakeholders

- Hostel management
- Students (room occupants)
- Maintenance staff
- Project development team

KPIs (Key Performance Indicators)

- Percentage reduction in electricity consumption
- Monthly electricity bill savings

- System reliability and uptime
 - Number of rooms successfully automated
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8. Conclusion

Summary of Findings

The Smart Hostel Electricity Saver Auto Cut-Off System provides an effective solution to reduce electricity wastage in hostel environments. By using sensors and microcontroller based automation, the system can detect room occupancy and automatically control the electrical supply.

Long-Term Implications

Implementing such systems can significantly contribute to:

- Energy conservation
- Reduced electricity costs
- Sustainable hostel infrastructure
- Adoption of smart building technologies

Final Remarks

The project demonstrates how simple automation technologies can address real-world energy management challenges. With proper implementation and maintenance, the system has the potential to become a practical solution for hostels, hotels, and residential buildings.

10. References

Academic Sources

- Energy Efficiency and Smart Building Technologies – IEEE Journals
- Smart Energy Management Systems – International Journal of Engineering Research

Books

- Microcontroller Based System Design – Raj Kamal

- Embedded Systems Architecture – Tammy Noergaard

Industry Sources

- Energy conservation guidelines from the International Energy Agency
- Smart building automation resources from the Institute of Electrical and Electronics Engineers